Introduction

Explore core data extraction techniques from relational and non-relational databases, APIs, web scraping, and cloud repositories. Understand different data formats and how to handle structured, semi-structured, and unstructured data. This lesson prepares you to build versatile ETL pipelines by mastering extraction from diverse data sources.

In today's world, data is everywhere. It’s constantly generated and stored in various sources. To effectively transfer data, it’s essential to first learn how to extract it.

This section introduces a range of techniques for extracting data from diverse sources such as relational and non-relational databases, cloud data warehouses, APIs, web scraping, and more. These skills will help us build ETL pipelines capable of extracting data from various sources in different formats and for multiple purposes.

Data sources

The method for extracting data from each source varies depending on the specific characteristics of the data, the source itself, and the purpose for which we extract the data.

Some common data sources include:

- **Relational databases:** They store data in structured tables and are simple to extract. Examples include MySQL, Oracle, and PostgreSQL.
- **Non-relational databases:** They store data as documents, key-value pairs, columns, or graphs. Examples include MongoDB, Neo4j, and Apache

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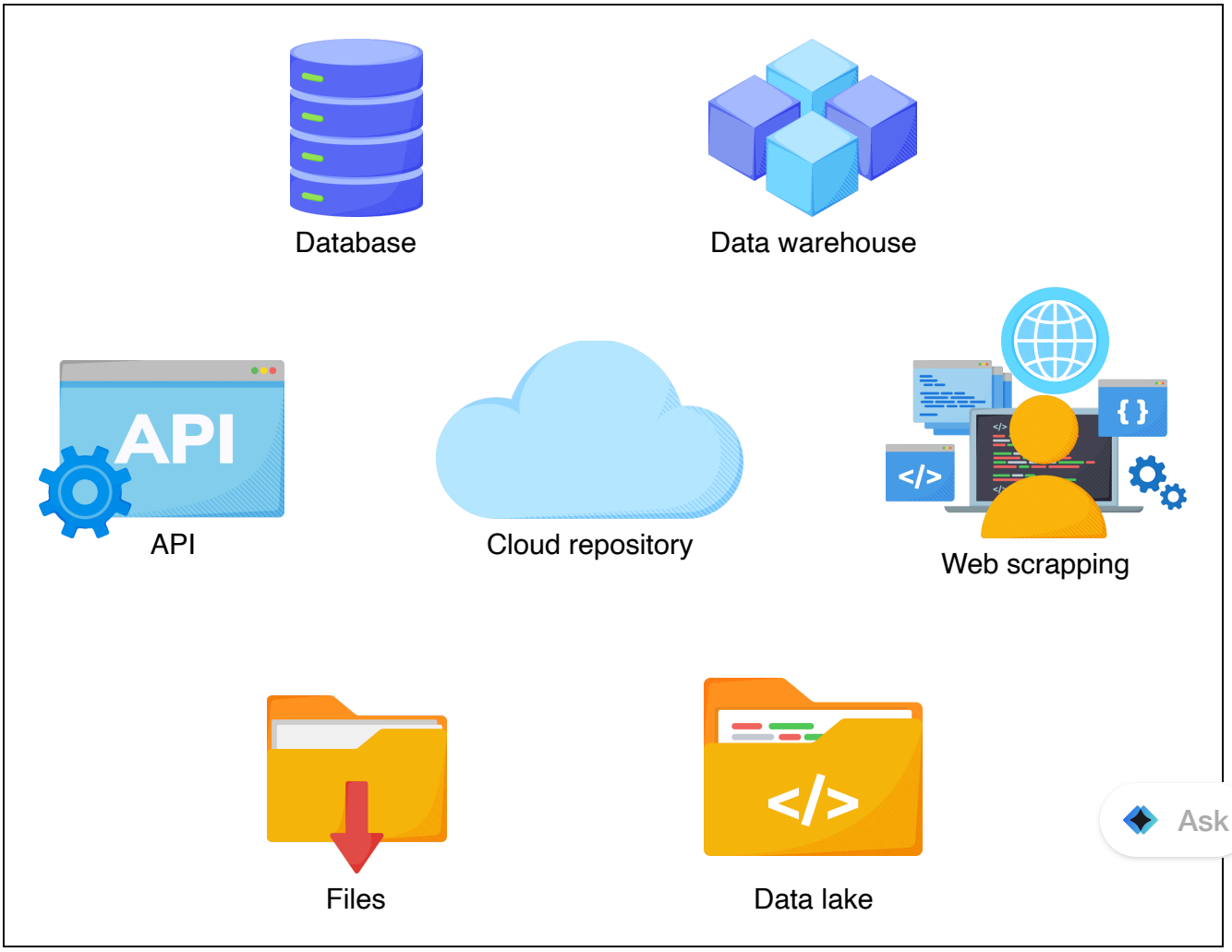
Cassandra.

- **Data warehouses:** They serve as central sources for business users and store structured data. Examples include Snowflake and Google BigQuery.
- **Data lakes:** They store structured, semi-structured, and unstructured data at a large scale. Examples include
- **APIs:** They can provide quick data access from web-based apps and online services.
- **Web scraping tools:** They're used to extract HTML content from web pages.
- **Streaming sources:** Only the ones that generate real-time data and require specialized handling. Examples include social media feeds.
- **Cloud repositories:** They store vast amounts of data and are commonly used for extraction.



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Data sources

Other than these sources, data can also be extracted from files and directories, downloaded manually, from a UI, or using a Bash command like `curl` or `wget`, or even generated using a programming language.

Data formats

One of the common hurdles in extracting data is that data is stored in various formats. Each format has its own strengths and weaknesses. Some are suited for analytics and are easy to deal with, such as tabulated data in CSV files, while others are suited for efficient compression and storage. Some common data formats are:

- **Structured and semi-structured data formats:** These data formats are easy to work with, have great readability, and are compatible with multiple software. However, they can be inflexible and inefficient for storing large volumes of data. Examples include CSV, JSON, XML, and Apache Parquet.
- **Unstructured data formats:** These data formats can be hard to work with because they don't have a predefined structure. However, this makes them flexible and well-suited for storing and sharing information. Examples include text format and PDF.
- **Binary data formats:** These store data in a binary format that's not human-readable and they require specialized software and techniques to process. However, they're well-suited for storing and transmitting large amounts of data because they're compact and efficient. Furthermore, they're widely supported and easily accessed by most software applications. Examples include image and video formats.

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Mark As Completed